Micro Commercial Components

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2N3904

NPN General Purpose Amplifier

Features

- Lead Free Finish/RoHS Compliant ("P" Suffix designates RoHS Compliant. See ordering information)
- ∉ Capable of 625mW of Power Disspation and 200mA Ic
- ∉ Epoxy meets UL 94 V-0 flammability rating
- ∉ Moisure Sensitivity Level 1
- ∉ Through Hole Package
- ∉ Halogen free available upon request by adding suffix "-HF"
- Operating Temperature: -55°C to +150°C
- Storage Temperature: -55°C to +150°C

Electrical Characteristics @ 25°C Unless Otherwise Specified

Symbol	Parameter	Min	Max	Units
OFF CHARACTERISTICS				
$V_{(BR)CEO}$	Collector-Emitter Breakdown Voltage* (I_C=1.0mAdc, I_B=0)	40		Vdc
$V_{(BR)CBO}$	Collector-Base Breakdown Voltage (I_C=10μAdc, I_E=0)	60		Vdc
$V_{(BR)EBO}$	Emitter-Base Breakdown Voltage (I_E=10μAdc, I_C=0)	6.0		Vdc
I_{BL}	Base Cutoff Current (V_{CE}=30Vdc, V_{BE}=3.0Vdc)		50	nAdc
I_{CEX}	Collector Cutoff Current (V_{CE}=30Vdc, V_{BE}=3.0Vdc)		50	nAdc
ON CHARACTERISTICS				
h_{FE}	DC Current Gain*			
	(I_C=0.1mAdc, V_{CE}=1.0Vdc)	40		
	(I_C=1.0mAdc, V_{CE}=1.0Vdc)	70		
	(I_C=10mAdc, V_{CE}=1.0Vdc)	100	300	
	(I_C=50mAdc, V_{CE}=1.0Vdc)	60		
	(I_C=100mAdc, V_{CE}=1.0Vdc)	30		
$V_{CE(sat)}$	Collector-Emitter Saturation Voltage			
	(I_C=10mAdc, I_B=1.0mAdc)		0.2	Vdc
	(I_C=50mAdc, I_B=5.0mAdc)		0.4	
$V_{BE(sat)}$	Base-Emitter Saturation Voltage			
	(I_C=10mAdc, I_B=1.0mAdc)	0.65	0.85	Vdc
	(I_C=50mAdc, I_B=5.0mAdc)		0.95	
SMALL-SIGNAL CHARACTERISTICS				
f_T	Current Gain-Bandwidth Product (I_C=10mAdc, V_{CE}=20Vdc, f=100MHz)	250		MHz
C_{obo}	Output Capacitance (V_{CB}=5.0Vdec, I_E=0, f=1.0MHz)		4.0	pF
C_{ibo}	Input Capacitance (V_{BE}=0.5Vdc, I_C=0, f=1.0MHz)		8.0	pF
NF	Noise Figure (I_C=100μAdc, V_{CE}=5.0Vdc, R_S=1.0kΩ f=10Hz to 15.7kHz)		5.0	dB

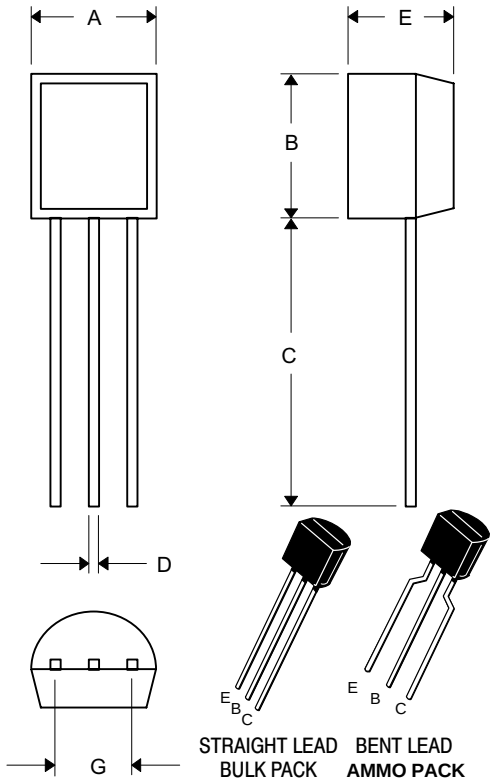
SWITCHING CHARACTERISTICS

t_d	Delay Time	(V_{CC}=3.0Vdc, V_{BE}=0.5Vdc	35	ns
t_r	Rise Time	I_C=10mAdc, I_{B1}=1.0mAdc)	35	ns
t_s	Storage Time	(V_{CC}=3.0Vdc, I_C=10mAdc	200	ns
t_f	Fall Time	I_{B1}=I_{B2}=1.0mAdc)	50	ns

*Pulse Width ≤ 300μs, Duty Cycle ≤ 2.0%

Symbol	Characteristic	Max	Unit
$R_{\theta JA}$	Thermal Resistance, Junction–to–Ambient	200	°C/W

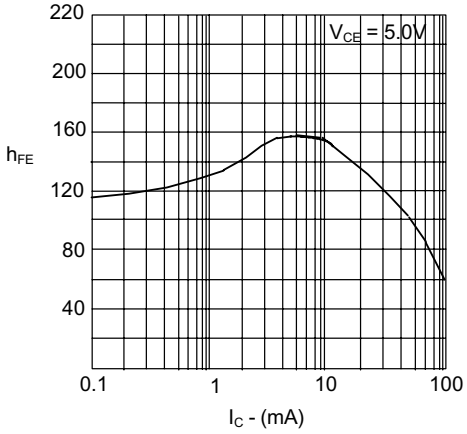
TO-92



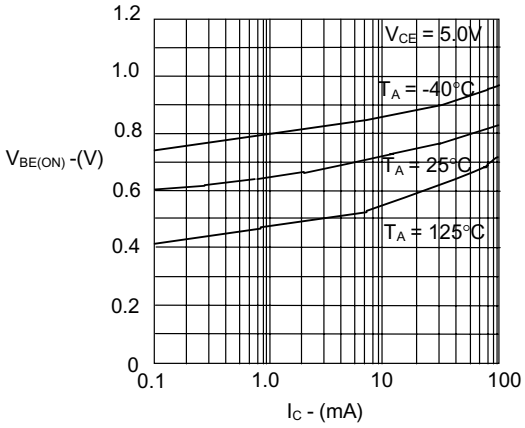
DIMENSIONS					
	INCHES		MM		
DIM	MIN	MAX	MIN	MAX	NOTE
A	.175	.185	4.45	4.70	
B	.175	.185	4.45	4.70	
C	.500	---	12.70	---	
D	.016	.020	0.41	0.63	
E	.135	.145	3.43	3.68	
G	.095	.105	2.42	2.67	Straight Lead
	.173	.220	4.40	5.60	Bent Lead

* For ammo packing detailed specification, click here to visit our website of product packaging for details.

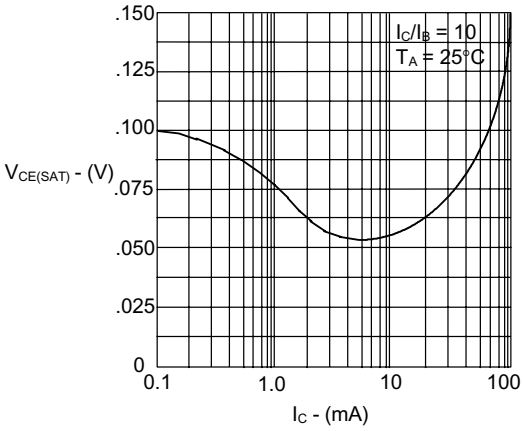
DC Current Gain vs Collector Current



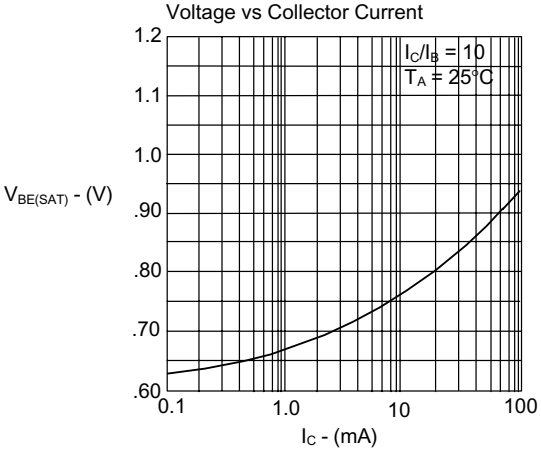
Base-Emitter ON Voltage vs Collector Current



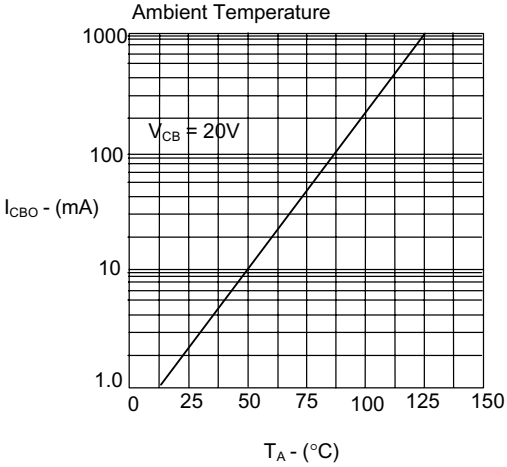
Collector Saturation Volatge vs Collector Current



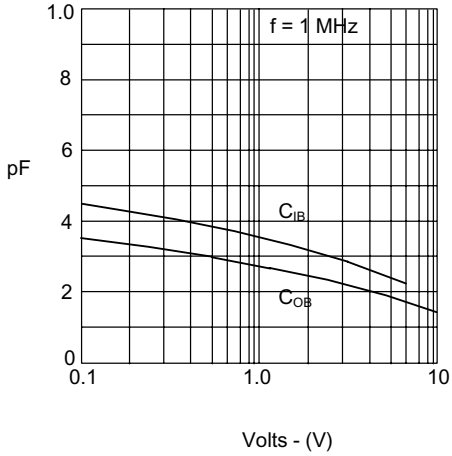
Base Saturation Voltage vs Collector Current



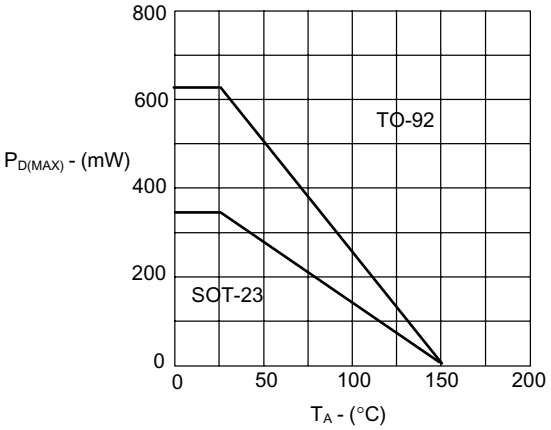
Collector Cutoff Current vs Ambient Temperature



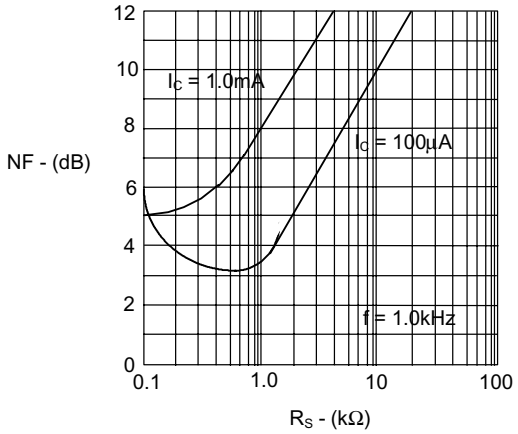
Capacitance vs Reverse Bias Voltage



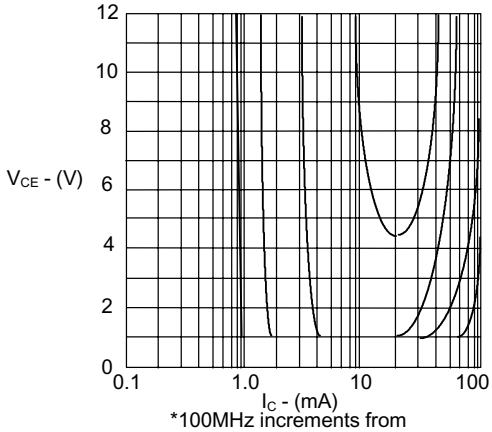
Maximum Power Dissipation vs Ambient Temperature



Noise Figure vs Source Resistance

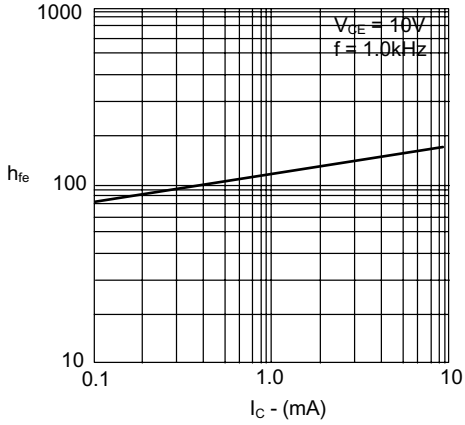


Contours of Constant Gain Bandwidth Product (f_T)

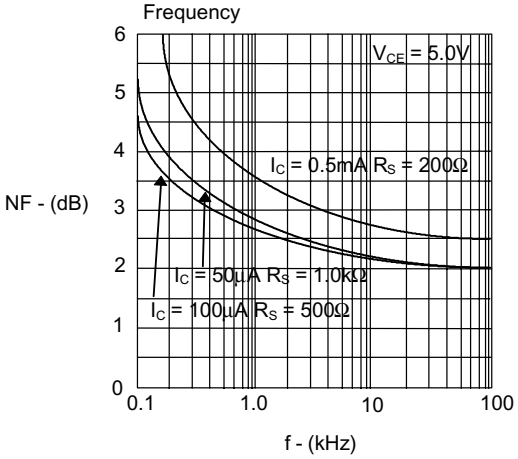


*100MHz increments from 200 to 500MHz

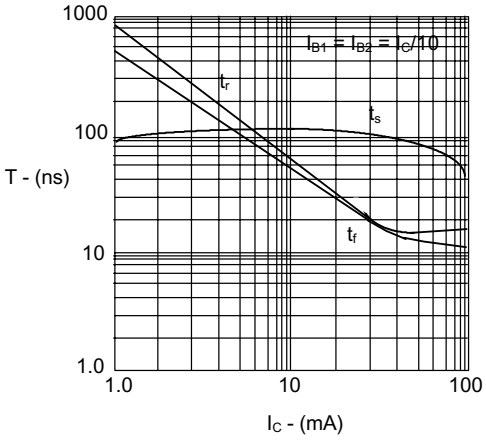
Current Gain

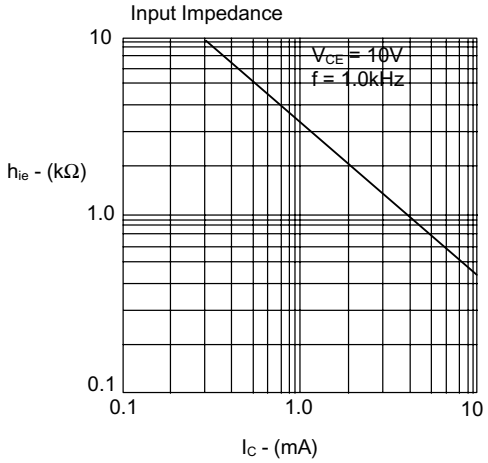


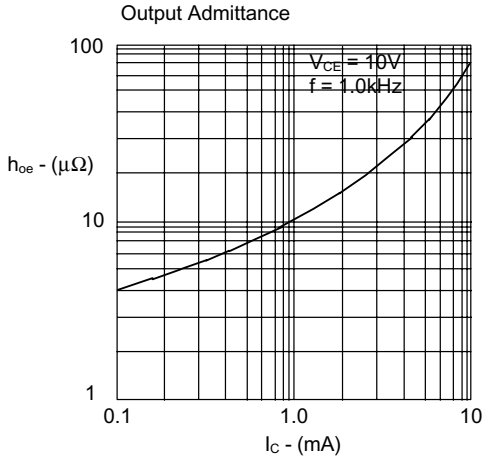
Noise Figure vs Frequency

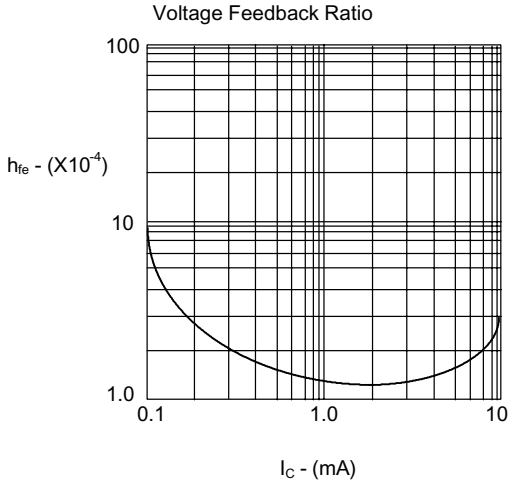


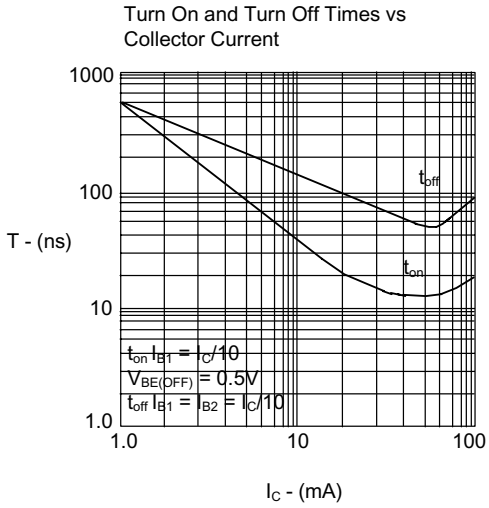
Switching Times vs Collector Current




Input Impedance
10
1.0
0.1
0.1
1.0
10
h_{ie} - (kΩ)
I_C - (mA)
V_{CE} = 10V
f = 1.0kHz


Output Admittance
100
10
1
0.1
1.0
10
h_{oe} - (μΩ)
I_C - (mA)
V_{CE} = 10V
f = 1.0kHz


Voltage Feedback Ratio
100
10
1.0
0.1
1.0
10
h_{fe} - ($X10^{-4}$)
I_C - (mA)


Turn On and Turn Off Times vs
Collector Current
1000
100
10
1.0
1.0
10
100
T - (ns)
I_C - (mA)
t_{off}
t_{on}
t_{on} I_{B1} = I_C/10
$V_{BE(OFF)}$ = 0.5V
t_{off} I_{B1} = I_{B2} = I_C/10

Ordering Information :

Device	Packing
Part Number-AP	Ammo Packing: 20Kpcs/Carton
Part Number-BP	Bulk: 100Kpcs/Carton

Note : Adding "-HF" suffix for halogen free, eg. Part Number-AP-HF

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